



# A panoramic review of IL-6: Structure, pathophysiological roles and inhibitors

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## ABSTRACT

Interleukin-6 (IL-6) is a pleiotropic pro-inflammatory cytokine. Its deregulation is associated with chronic inflammation, and multifactorial auto-immune disorders. It mediates its biological roles through a hexameric complex composed of IL-6 itself, its receptor IL-6R, and glycoprotein 130 (IL-6/IL-6R/gp130). This complex, in turn, activates different signaling mechanisms (classical and trans-signaling) to execute various biochemical functions. The trans-signaling mechanism activates various pathological routes, like JAK/STAT3, Ras/MAPK, PI3K-PKB/Akt, and regulation of CD4+ T cells and VEGF levels, which cause cancer, multiple sclerosis, rheumatoid arthritis, anemia, inflammatory bowel disease, Crohn's disease, and Alzheimer's disease. Involvement of IL-6 in pathophysiology of these complex diseases makes it an important target for the treatment of these diseases. Though some anti-IL-6 monoclonal antibodies are being used clinically, but their high cost, only parenteral administration, and possibility of immunogenicity have limited their use, and warranted the development of novel small non-peptide molecules as IL-6 inhibitors. In the present report, all molecules reported in literature as IL-6 inhibitors have been classified as IL-6 production, IL-6R, and IL-6 signaling inhibitors. Reports available till date are critically studied to identify important and salient structural features common in these molecules. These analyses would assist medicinal chemists to design novel and potent IL-6 production and signaling inhibitors, through knowledge- and/or computer-based approaches, for the treatment of complex multifactorial diseases.

## 1. Introduction

Inflammation is a natural defence mechanism of the body that involves migration of leucocytes to the damaged tissues to destroy the inflammatory trigger or challenge. Infectious and allergic triggers induce an initial phase of acute inflammation, which is considered to have limited beneficial effects. But, the continued acute inflammation leads to chronic inflammation that culminates into tissue damage. Acute inflammation is characterized by infiltration of neutrophilic cells followed by monocytic cells, whereas chronic inflammation is characterized by the presence of mononuclear cells such as macrophages and lymphocytes at the site of inflammation.<sup>1–3</sup> In addition to the recruitment of these cells at the inflammatory site, various cytokines are also produced during the inflammatory processes. These act in synergy and have the overlapping activity that may occur either at their receptor or along their intracellular signal transduction pathways. These cytokines mainly include interleukin-6 (IL-6), interleukin-1 $\beta$  (IL-1 $\beta$ ), tumor necrosis factor- $\alpha$  (TNF- $\alpha$ ), interferon- $\gamma$  (IFN- $\gamma$ ) and transforming

growth factor- $\beta$  (TGF- $\beta$ ), which stimulate the production of acute phase proteins (APPs).<sup>4</sup> IL-6 is the one that not only elicits acute phase reactions but also leads to the development of specific cellular and humoral immune responses *via* end-stage B-cell differentiation, immunoglobulin secretion and T-cell activation. Thus, IL-6 is an important modulator for the transition from acute phase to chronic phase of inflammation.<sup>5</sup>

IL-6 is a multifunctional cytokine that plays both pro-inflammatory and anti-inflammatory roles in humans.<sup>6</sup> Conventionally, it is known with diverse names depending on their biological roles and sources. It was called as interferon- $\beta_2$  for its cloning to isolate and characterize viral-induced protein interferon- $\beta$ .<sup>7,8</sup> Thereafter, it has been described as 26 K factor,<sup>9,10</sup> B-cell stimulatory factor 2,<sup>11</sup> hybridoma growth factor,<sup>12,13</sup> plasmacytoma growth factor,<sup>14</sup> hepatocytestimulatory factor,<sup>15</sup> a hematopoietic factor,<sup>16</sup> and cytotoxic T-cell differentiation factor.<sup>17</sup> Way back in 1993, William Ershler indicated IL-6 as the main modulator in signaling pathway in aging and chronic morbidity, and this statement turned out to be prophetic.<sup>18</sup> It plays a central role in

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diverse host defence mechanisms such as immune response, haematopoiesis, and acute-phase reactions, and thus, implicated in the pathology of many immuno-inflammatory diseases. The other IL-6 type cytokines include IL-11, leukaemia inhibitory factor (LIF), oncostatin M (OSM), ciliary neurotrophic factor (CNTF), cardiotrophin-1 (CT-1) and cardiotrophin-like cytokine (CLC) that activate target genes involved in the differentiation, survival, apoptosis and proliferation of cells, depending upon the nature of target cells.<sup>19</sup> In general, the cytokines vary in their receptor systems *i.e.*, low affinity ligand binding  $\alpha$ -chain and signal transducing high affinity converting  $\beta$ -chain. The family of IL-6 type cytokines specifically share a  $\beta$ -chain, called as gp130 (glycoprotein 130), which is responsible for their functional activities through various mechanisms.<sup>20</sup>

## 2. Structures of IL-6 and its receptor complex

Detailed studies on structures of IL-6, its receptor (IL-6R) and the IL-6-receptor complex are available in literature.<sup>21–28</sup> Nevertheless a brief description of these is required for the understanding of medicinal chemists. IL-6 is a single chain phosphorylated glycoprotein consisting of mainly four helix bundles (A–D). The helices A and B run in one direction whereas C and D run in the opposite direction, and their representation in ribbon form is shown in Fig. 1a. Helix E is the other short chain between helices C and D that lies outside the main four-helix bundle (A–D). IL-6 transmits its signals through a cell-surface type-I receptor complex, that consists of a ligand-binding glycoprotein termed as IL-6 receptor (IL-6R) and a signal-transducing component gp130. IL-6R is a  $\alpha$ -chain (80 kDa) and is also called as CD126. It consists of three domains *i.e.*, D<sub>1</sub>, D<sub>2</sub> and D<sub>3</sub> (Fig. 1b).<sup>25</sup> The D<sub>1</sub> is also known as Ig domain, and is linked to CBD (cytokine binding domain), which actually is comprised of D<sub>2</sub> and D<sub>3</sub> domains. IL-6R exists in two forms *i.e.*, membrane bound (mIL-6R) and soluble (sIL-6R). mIL-6R is expressed mainly in hepatocytes, neutrophils, monocytes, and T-cells, and executes IL-6-mediated normal physiological functions such as induction of APP in liver, effects on neutrophils and acquired immunity.<sup>23</sup> sIL-6R is produced from mIL-6R by proteolytic cleavage of metalloproteinases gene family members (ADAM10 and ADAM17), or by

alternative splicing of IL-6R mRNA.<sup>24</sup> It binds to IL-6 with low affinity to form a dimer (IL-6-sIL-6R complex) that travels throughout the body. This dimer binds to gp130 to form a heterotrimer (Fig. 1d), which further binds with another heterotrimer to give a hexameric complex (Fig. 1e). This hexameric complex executes varied pathological functions depending upon the location of gp130. Therefore, sIL-6R acts as a carrier of IL-6 to the pathological site containing gp130. gp130 is the tallest cytokine receptor and a  $\beta$ -chain (130 kDa), which is also called as CD130.<sup>21</sup> The N-terminal residue of gp130 constitutes its extracellular portion, and is composed of six domains (D<sub>1</sub>–D<sub>6</sub>). Similarly as in IL-6R, D<sub>1</sub> is Ig domain, and D<sub>2</sub> and D<sub>3</sub> constitute the CBD. The other three domains (D<sub>4</sub>–D<sub>6</sub>) are important components for binding with IL-6, and for stimulating the subsequent signaling mechanisms (Fig. 1c).

## 3. Signaling mechanisms of IL-6

IL-6 transmits signal through mIL-6R (classical signaling pathway) or through sIL-6R (trans-signaling pathway) (Fig. 2). In both the cases, IL-6 first binds to the receptor and then to gp130 *via* CBD domains, but elicits different biological effects depending upon the receptor form.<sup>29</sup> Classic signaling is manifested mainly in leukocytes and liver cells, which express both mIL-6R $\alpha$  and gp130, and promotes the anti-inflammatory responses. In contrast, trans-signaling can be manifested in all gp130-expressing cells, and leads to pro-inflammatory responses.<sup>30</sup> These pro- and anti-inflammatory responses are elicited by a hexameric complex of IL-6 with IL-6R and gp130, as discussed in the preceding section. This hexameric complex activates Janus kinase (JAK)<sup>28</sup> that, in turn, activates three possible signaling routes. In route 1, JAK induces tyrosine phosphorylation of itself, and subsequently activates dimerization of signal transducer and transcription-3 (STAT3).<sup>19</sup> In route 2, JAK activates Ras/Raf pathway, which subsequently causes hyperphosphorylation of mitogen activated protein kinases (MAPK) and an increase in its serine/threonine kinase activity.<sup>29</sup> The route 3 involves the activation of phosphoinositol-3 kinase (PI3K) – protein kinase B (PKB)/Akt pathway, wherein JAK phosphorylates and activate PI3K, which then phosphorylates certain phosphatidylinositides to phosphatidylinositol-4,5-bisphosphate (PIP2) and phosphatidylinositol-3,4,5-

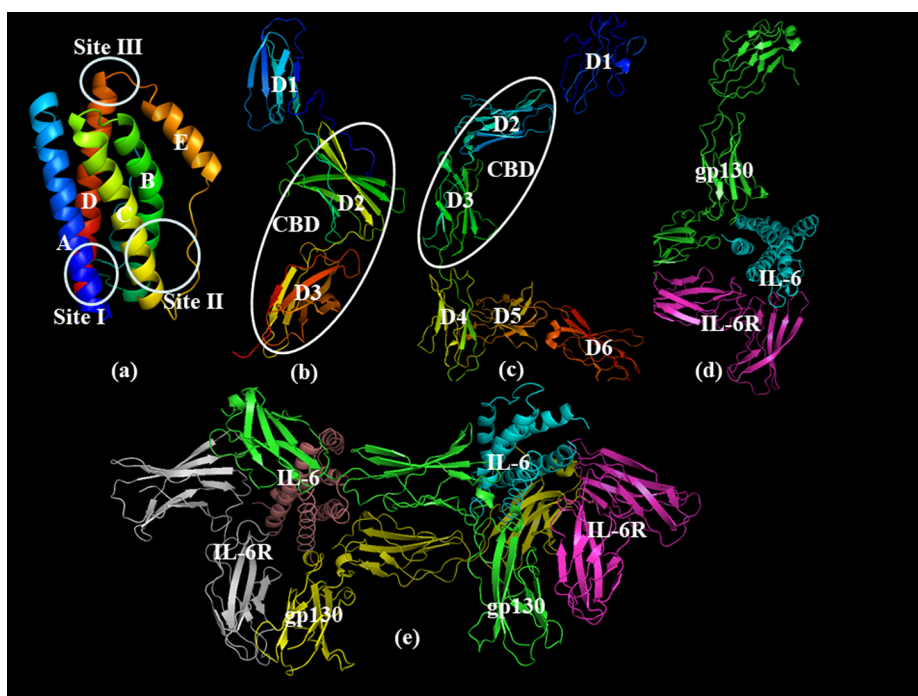


Fig. 1. Ribbon structure representation of IL-6 and its receptor: (a) Structure of IL-6; (b) Structure of IL-6R; (c) Structure of gp130; (d) Heterotrimer of IL-6, IL-6R and gp130; (e) Hexameric form of receptor complex.

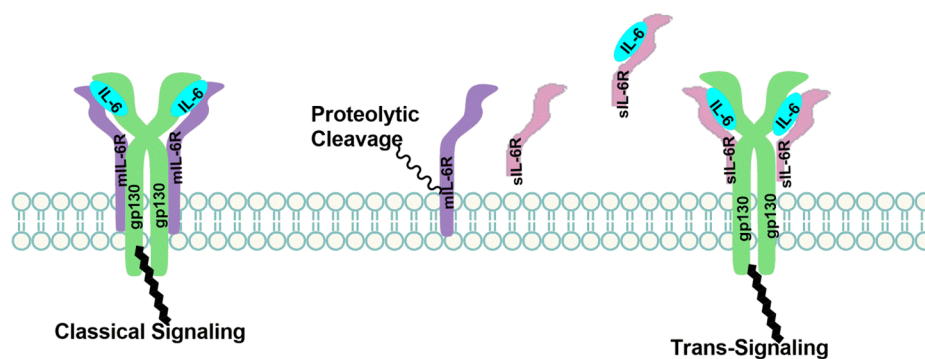


Fig. 2. Signaling mechanisms of IL-6.

triphosphate (PIP3). PIP3, in turn, phosphorylates and activates PKB/Akt, which is recruited to plasma membrane.<sup>31</sup> This PI3K/Akt pathway contributes to NF- $\kappa$ B activation.<sup>32</sup> Activation of STAT3, MAPK, PKB/Akt pathways and NF- $\kappa$ B leads to varied pathological conditions such as cancer, multiple sclerosis (MS), rheumatoid arthritis (RA), diabetes, anemia, inflammatory bowel disease (IBD), Crohn's disease (CD), and Alzheimer's disease (AD). Thus, IL-6 is implicated in both specific physiological and pathological functions depending upon its biochemical pathways.

#### 4. Physiological roles of IL-6

IL-6 participates actively in inflammatory and immunomodulatory mechanisms. It stimulates the production of APPs that induce acute phase reactions,<sup>33</sup> acts as a maturing agent for B-lymphocytes,<sup>34</sup> stimulates the synthesis and secretion of various immunoglobulins,<sup>35</sup> and induces the proliferation of thymic and peripheral T-cells.<sup>36</sup> In combination with IL-1, it causes differentiation of T-cells into cytolytic T-cells<sup>37</sup> that further activate natural killer cells.<sup>38</sup> These killer cells participate in causative mechanisms that are responsible for specific diseases associated with the inflammatory or immune system alternations. In reproductive system, IL-6 participates in menses<sup>39</sup> and spermatogenesis.<sup>40</sup> IL-6 also plays an important role in bone metabolism through induction of osteoclastogenesis and osteoclast activity.<sup>41,42</sup> In rodents, inhibition of IL-6 gene expression is, in part, responsible for the inhibition of osteoclast activation.<sup>43,44</sup> This role of IL-6 in bone strengthening is supported by another report wherein IL-6 gene-knockout mice are found protected from cancellous bone loss associated with ovariectomy.<sup>45</sup> IL-6 has a beneficial impact in numerous liver pathologies due to its role in liver regeneration and in promoting an anti-inflammatory response in certain conditions. In addition to these activities, it also plays functional roles in a variety of other processes including skin proliferation,<sup>46</sup> megakaryocytopoiesis,<sup>47</sup> macrophage differentiation,<sup>48,49</sup> and neural cell differentiation and proliferation.<sup>50,51</sup>

#### 5. Pathological roles of IL-6

##### 5.1. In cancer

IL-6 is the major cytokine in tumor microenvironment, and is known to be deregulated in cancer. It is over-expressed in almost all types of tumors such as breast cancer,<sup>52</sup> prostate cancer,<sup>53</sup> ovarian carcinoma,<sup>54</sup> pancreatic cancer,<sup>55</sup> lung cancer,<sup>56</sup> renal cell carcinoma,<sup>57</sup> cervical cancer<sup>58</sup> and multiple myeloma.<sup>59</sup> The three signaling pathways i.e., JAK/STAT3, Ras/MAPK and PI3K–PKB/Akt, initiated by IL-6, regulate many gene products that cause cell proliferation, differentiation, apoptosis, angiogenesis and metastasis (Fig. 3).<sup>60</sup> STAT3-mediated proliferation is caused via inhibition of various apoptotic genes such as Bcl-2, Bcl-xl, survivin, induced Mcl-1 and XIAP.<sup>61,62</sup> The activated

STAT-3 up-regulates the expression of bFGF and Matrix metalloproteinase-9 (MMP-9) in tumor-associated myeloid cells and endothelial cells that contribute to tumor angiogenesis.<sup>63</sup>

The activation of STAT3 and MAPK affects angiogenesis by inducing VEGF expression in tumor-associated endothelial cells.<sup>64</sup> The activated PI3K–PKB/Akt promotes cell proliferation and cell survival through the inhibition of p53,<sup>65</sup> induction of cyclin D1 and myc, inactivation of Chk-1, and transcription of tumor suppressors p27Kip1 and p130Rb2.<sup>66</sup> PI3K/Akt also increases protein synthesis and cell growth via activation of mTOR, a Ser/Thr kinase that regulates cell growth and metabolism.<sup>67</sup> Moreover, PI3K/Akt pathway contributes to NF- $\kappa$ B activation leading to increased cell survival and regulation of angiogenesis and invasiveness.<sup>68</sup> Many clinical studies on the use of anti-IL-6 monoclonal antibodies (mAbs, BE-8 or Cnto 328) in patients suffering from multiple myeloma, renal cell carcinoma, and B-lymphoproliferative disorders have proved these mAbs to be beneficial in all patients. These literature reports suggest that IL-6 therapy may be useful in treating cancer at an earlier stage.<sup>69,70</sup>

##### 5.2. In multiple sclerosis

It is an inflammatory autoimmune disease with an unknown etiology,<sup>71</sup> wherein insulating myelin sheaths are successively destroyed leading to an array of neuropathological symptoms including pain, mental, and sometimes psychiatric problems.<sup>72</sup> There are evidences of increased levels of IL-6 in cerebrospinal fluid (CSF) and plasma of patients suffering from multiple sclerosis.<sup>73,74</sup> Studies have also suggested that neuropathological manifestations, including the degeneration of grey and white matter,<sup>75</sup> are induced by action of IL-6 on astrocytes and glial cell in brain.<sup>76</sup> Mechanistically, this neuropathological role of IL-6 is due to its binding with sIL-6 that causes an increase in the expression of Th17-cells, which ultimately leads to demyelination of axon.<sup>77</sup>

##### 5.3. In Alzheimer's disease (AD)

AD is a progressive neurodegenerative disorder of brain, characterised by the formation of neurofibrillary tangles (NFTs) and  $\beta$ -amyloid plaques (A $\beta$ ). IL-6 plays a central pathogenic role in chronic stress and physiologic aging that increase the propensity of formation of NFTs and A $\beta$ .<sup>32</sup> JAK tyrosine kinase, activated through the hexameric complex, activates dimerization of STAT3.<sup>19</sup> It, in turn, activates Ras protein, which causes hyperphosphorylation of MAPK and an increase in its serine/threonine kinase activity (Fig. 3).<sup>78</sup> This activation of JAK/STAT3 and MAPK leads to hyperphosphorylation of tau protein, which is responsible for the formation of NFTs.<sup>79</sup> JAK also contributes to the activation of NF- $\kappa$ B via PI3K/Akt pathway. NF- $\kappa$ B, in turn, binds with amyloid precursor protein promoter and synthesizes A $\beta$  leading to the formation of senile plaques.<sup>32</sup>

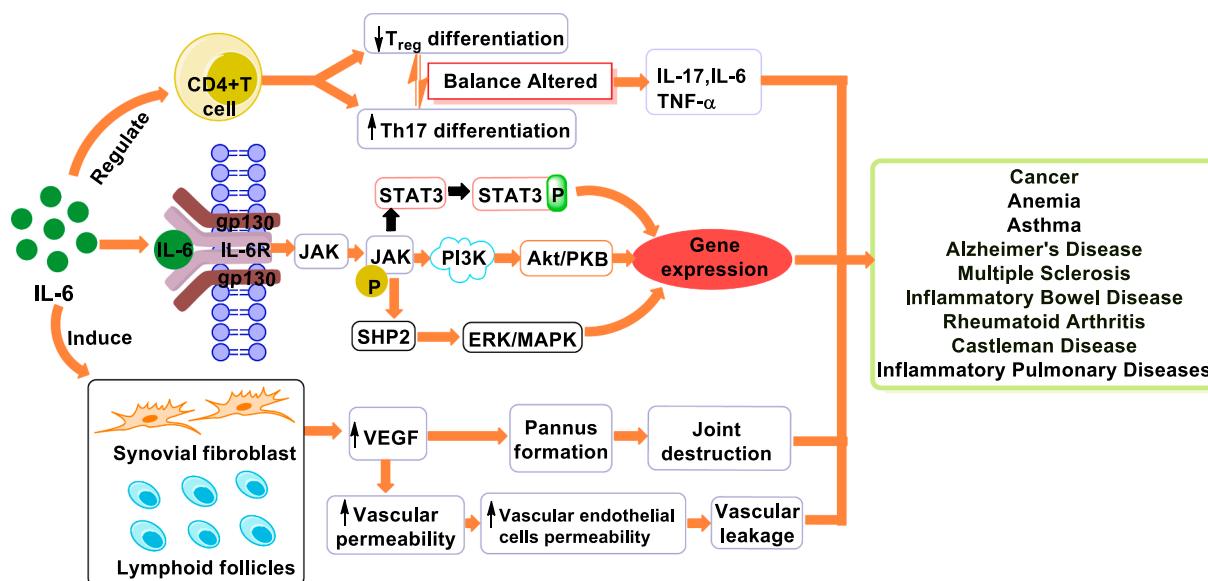


Fig. 3. IL-6-induced pathways in various immunoinflammatory diseases.

#### 5.4. In Rheumatoid arthritis (RA) and castleman disease (CD)

Both CD and RA are characterized by systemic inflammation with elevated levels of serum IL-6. CD is a chronic lympho-proliferative disorder, mainly caused by overproduction of IL-6.<sup>80,81</sup> It is not accompanied by destructive arthritis, and that makes it different from RA. In CD, IL-6 is produced by lymph nodes but in RA, it is produced by synovium. It is expressed in joints of patients in very early stages of RA<sup>82</sup> and promotes synovitis and joint destruction by stimulating neutrophil migration and osteoclast maturation. IL-6/sIL-6R complex also increases the expression of VEGF and bone resorption, which further leads to pannus formation.<sup>83</sup> Inhibition of IL-6 through mAbs is useful in the treatment of RA and CD. Tocilizumab (a human anti-IL-6R), siltuximab (an IL-6 inhibitor) and tofacitinib (a JAK inhibitor) are the approved mAbs that are used for the treatment of CD, RA and systemic juvenile idiopathic.<sup>84,85</sup>

#### 5.5. In anemia

Hepcidin is an APP responsible for regulation of intestinal iron absorption, iron recycling by macrophages, and iron metabolism. Over expression of IL-6 in transgenic mice increases plasma hepcidin concentration that results in anemia.<sup>86</sup> IL-6 binds to its cellular receptor and activates STAT3. The activated STAT3 binds to hepcidin promoter, which increases the level of hepcidin. IL-6 is reported to be the sole inducer of hepcidin mRNA, which mediates inflammation.<sup>87</sup>

#### 5.6. In inflammatory bowel disease (IBD)

IBD, a chronic inflammation of gastrointestinal tract, mainly includes Crohn's disease and Ulcerative colites.<sup>88</sup> IL-6 plays a crucial role in IBD via uncontrolled intestinal inflammatory process.<sup>89</sup> Serum and intestinal levels of IL-6 and sIL-6R increase significantly in Crohn's disease,<sup>90,91</sup> and there is a direct relation between IL-6 levels and inflammation.<sup>92</sup> The elevated level of IL-6 contributes to an enhanced survival of T-cells and apoptosis resistance in lamina propria at the inflamed site. This expansion of CD4+ T cells leads to perpetuation of inflammation. IL-6 also elicits important homeostatic functions by stimulating the proliferation and expansion of intestinal epithelial cells (IECs). The blocking of IL-6 signaling with mAbs suppresses chronic intestinal inflammation effectively, which implicates IL-6 as a potential therapeutic target in IBD.<sup>93</sup> A clinical trial on tocilizumab, in patients

with Crohn's disease has revealed that levels of acute-phase reactants are completely normalized after a single dose of tocilizumab.<sup>94</sup> This striking improvement in the acute-phase reactants confirms the involvement of IL-6 as major cytokine in Crohn's disease.

#### 5.7. In asthma and inflammatory pulmonary diseases

IL-6, in lungs, is produced from epithelial cells, interstitial fibroblasts, macrophages and other inflammatory cells in response to a variety of stimuli that include allergens, respiratory viruses, exercise, environmental particles, and inhaled toxic particles. Herein, it regulates the CD4+ T cell-mediated responses that include cytokine production (IL-4, IL-13, IL-17 and IL-21), sIL-6R production, and suppressive activity on Treg cells. These mediators or responses, in turn, contribute to damage lungs through their effects on mucus production, matrix deposition, and proteases release from granulocytes among others.<sup>95–98</sup> An Australian genome wide association study on more than 2000 asthmatic and 4000 control individuals has revealed that one single nucleotide polymorphism (SNP, rs4129267) in intron-8 of IL-6R gene in chromosome 1q21.3 is associated with asthma risk. This polymorphic form of IL-6 SNP is also reported to be associated with chronic obstructive pulmonary disease (COPD). The study has suggested that SNP IL-6 does not affect the serum level of IL-6, and its association with COPD may be due to the effect of IL-6 variants on the genes that affect other mechanism(s) of COPD.<sup>99</sup>

### 6. IL-6 inhibitors

Anti-IL-6 mAbs are used to treat various immunoinflammatory diseases that are refractory to conventional drugs. Tocilizumab is a humanized anti-IL-6R mAb, which is approved by FDA for the treatment of RA and systemic juvenile idiopathic arthritis. It is also approved in Japan for the treatment of CD, and is currently in Phase 2 trials for assessment of its efficacy in relapsing polychondritis.<sup>100</sup> Clinical studies on anti-IL-6 mAbs such as sirukumab (CNT0136), olokizumab (CP6038), PF-423691, siltuximab (CNT0328), elsilimomab (BE-8), clazakizumab (BMS945429), sarilumab (REGN88), and MEDI5117 are in different phases of clinical trials to establish their efficacy and safety in varied disease states. The details of ongoing or completed trials on these anti-IL-6 mAbs are summarised in Table 1.<sup>101–110</sup> Many studies have revealed that these mAbs show beneficial therapeutic effects in IL-6-mediated diseased states. Some of



**Table 1**  
Various IL-6 mAbs and their clinical trials status.

| mAb                                                             | Indication                    | Status <sup>a</sup> /Govt. trial ID <sup>b</sup> | Sponsor                                   | Title                                                                                                                                                                                                                            | Remarks/Findings                                                                                                                                                                                        |
|-----------------------------------------------------------------|-------------------------------|--------------------------------------------------|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>IL-6 receptor inhibitors</b><br>Tocilizumab                  | Rheumatoid arthritis          | Completed/<br>NCT01119859                        | Hoffmann-La Roche                         | A study of tocilizumab (roactemra/actemra) versus adalimumab in patients with rheumatoid arthritis.                                                                                                                              | This study evaluated the efficacy and safety of tocilizumab versus adalimumab as monotherapy in patients with rheumatoid arthritis who are intolerant of methotrexate.                                  |
|                                                                 | Obesity                       | Not yet recruiting/<br>NCT03967691               | Rigshospitalet, Denmark                   | IL-6 regulation of substrate metabolism and influence of obesity.                                                                                                                                                                | Investigation of physiological role of IL-6 in lipid, glucose and protein metabolism during rest and exercise in healthy and obese humans.                                                              |
|                                                                 | Relapsing polychondritis      | Completed/<br>NCT01041248                        | Children's Hospital of Eastern Ontario    | Single patient study to treat relapsing polychondritis with tocilizumab.                                                                                                                                                         | Tocilizumab is effective in treatment of relapsing polychondritis at a dose of 8 mg/kg every two weeks. <sup>101</sup>                                                                                  |
|                                                                 | Myocardial Infarction         | Recruiting/<br>NCT030004703                      | Oslo University Hospital                  | Assessing the effect of anti-IL-6 treatment in myocardial infarction: The ASSAIL-MI trial (ASSAIL-MI).                                                                                                                           | Investigation on the ability of tocilizumab to reduce myocardial damage in patients presenting with an acute ST-segment elevation myocardial infarction.                                                |
|                                                                 | HIV Infection                 | Not Recruiting/<br>NCT02049437                   | Case Western Reserve University           | AIDS 347: IL-6 blockade in treated HIV infection.                                                                                                                                                                                | To investigate the effect of tocilizumab in HIV-infected subjects receiving antiretroviral therapy with suppressed viral replication.                                                                   |
|                                                                 | Cardiac Arrest                | Not Yet Recruiting/<br>NCT03863015               | Christian Hassager                        | IL-6 inhibition for modulating inflammation after cardiac arrest (IMICA).                                                                                                                                                        | To investigate the effect of systemic inflammation response in cardiac arrest patients using tocilizumab.                                                                                               |
|                                                                 | Ovarian Cancer                | Completed/<br>NCT01637532                        | Leiden University Medical Center          | Feasibility of the combination of chemotherapy (Carbo/ Caelyx or Carbo/ Doxorubicin) with tocilizumab (mAb IL-6R) and peg-intron in patients with recurrent ovarian cancer (PITCH).                                              | Blocking of functional IL-6R is feasible and safe in ovary cancer patients, treated with carboplatin/doxorubicin, by tocilizumab (8 mg/kg). Further recommended for Phase II evaluation. <sup>102</sup> |
| Sarilumab (REGN88, SAR153191)                                   | Solid Tumor Colitis Arthritis | Recruiting/<br>NCT03601611                       | Herlev Hospital                           | Checkpoint inhibitor induced colitis and arthritis -immunomodulation with IL-6 blockade and exploration of disease mechanisms (COLAR).                                                                                           | The study is aimed to investigate the expression of immunological genes in some disease with immune related adverse events, especially in tumor, colitis and arthritis. Results are not yet posted.     |
|                                                                 | Rheumatoid arthritis          | Completed/<br>NCT01055899                        | Regeneron Pharmaceuticals                 | Safety study of a single dose of REGN88 (SAR153191) in patients with RA.                                                                                                                                                         |                                                                                                                                                                                                         |
|                                                                 | Rheumatoid arthritis          | Completed/<br>NCT01011959                        | Regeneron Pharmaceuticals                 | A multiple-dose study of the safety and tolerability of REGN88 (SAR153191) in RA subjects.                                                                                                                                       | The study investigates the randomized study of the safety and tolerability of multiple doses of sarilumab in RA patients who are receiving treatment with methotrexate.                                 |
|                                                                 | Ankylosing spondylitis        | Completed/<br>NCT01061723                        | Sanofi Regeneron Pharmaceuticals          | Dose ranging study to evaluate the efficacy and safety of SAR153191 (REGN88) in patients with ankylosing spondylitis (ALIGN).                                                                                                    | Sarilumab (at doses of 100, 150 and 200 mg) is not effective in the treatment of ankylosing spondylitis. <sup>103</sup>                                                                                 |
|                                                                 | Rheumatoid arthritis          | Completed/<br>NCT01009242                        | UCB Pharma                                | To evaluate the blood levels and safety of IV and subcutaneous CDP6038 in subjects with RA using methotrexate.                                                                                                                   | The study determined the absolute bioavailability, safety, and relationship between blood levels of OKZ and suppression of C-reactive protein in single doses. The results are not yet posted.          |
| <b>IL-6 production inhibitors</b><br>Ollokizumab (OKZ, CDP6038) | Rheumatoid arthritis          | Completed/<br>NCT01296711                        | UCB BIOSCIENCES, Inc                      | Open-label study to assess the safety and efficacy of CDP6038 in patients who completed RA0056.                                                                                                                                  | Results are not yet posted.                                                                                                                                                                             |
|                                                                 | Rheumatoid arthritis          | Completed/<br>NCT01463059                        | UCB Japan Co. Ltd                         | Efficacy and safety of OKZ with RA with previously failed to anti-tumor necrosis factor (Anti-TNF) therapy.                                                                                                                      | The study evaluated the efficacy and safety of OKZ (administered at doses of 60, 120, 240 mg, s.c.) in comparison to placebo. The results are not yet posted.                                           |
|                                                                 | Rheumatoid arthritis          | Recruiting/<br>NCT03120949                       | R-Pharm                                   | Evaluation of the long term safety, tolerability and efficacy of two dosing regimens of OKZ, in subjects with RA who previously completed 24 weeks of blinded treatment in one of the core studies - CREDO 1, 2 or 3. (CREDO 4). | It is a Phase III study.                                                                                                                                                                                |
|                                                                 | Rheumatoid arthritis          | Recruiting/<br>NCT02760407                       | R-Pharm                                   | Evaluation of the effectiveness and safety of two dosing regimens of OKZ, compared to placebo and adalimumab, in subjects with RA who are taking methotrexate but have active disease.                                           | In Phase III recruiting the patients.                                                                                                                                                                   |
|                                                                 | Rheumatoid arthritis          | Completed/<br>NCT00718718                        | Centocor, Inc.                            | A study of effectiveness and safety of CNT0136 in patients with active RA despite methotrexate therapy.                                                                                                                          | CNT0136 showed improvements in the signs/symptoms in RA patients, and is safe. <sup>104</sup>                                                                                                           |
| Sirukumab (CNT0136)                                             | Lupus nephritis               | Completed/<br>NCT01273389                        | Janssen Research & Development, LLC (JRD) | An efficacy and safety study of CNT0136 in patients with active lupus nephritis.                                                                                                                                                 | CNT0136 is not found safe for patients with active lupus nephritis. <sup>105</sup>                                                                                                                      |

(continued on next page)

Table 1 (continued)

| mAb                      | Indication                                  | Status <sup>a</sup> /Govt. trial ID <sup>b</sup> | Sponsor                        | Title                                                                                                                                                             | Remarks/Findings                                                                                                                                                                          |
|--------------------------|---------------------------------------------|--------------------------------------------------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Siltuximab (CINTO328)    | Schizophrenia                               | Recruiting/<br>NCT02796859                       | Brian Miller                   | Siltuximab in schizophrenia.                                                                                                                                      | Investigation of safety, tolerability, and efficacy of siltuximab as an adjunct to antipsychotic medications in stable outpatients with schizophrenia.                                    |
|                          | Multiple myeloma                            | Completed/<br>NCT00911859                        | Janssen Research & Development | A study to compare CINTO328 and VELCADE-melphalan-prednisone (VMP) with VMP alone in previously untreated multiple myeloma patients.                              | Siltuximab to VMP does not improve the complete response rate in multiple myeloma. <sup>106</sup>                                                                                         |
| Clazakizumab (BMS945429) | Rheumatoid arthritis                        | Completed/<br>NCT01373151                        | Vitaeris INC                   | Phase IIB RA Dose ranging study for BMS-945429 in subjects who are not responding to methotrexate.                                                                | The clazakizumab treatment, alone or in combination with methotrexate, is well tolerated, and the patients show significant improvement. <sup>107</sup>                                   |
|                          | Non-Small Cell Lung Cancer-related Cachexia | Completed/<br>NCT00866970                        | Vitaeris INC                   | Safety, efficacy and pharmacokinetics of ALD518 in patients with non-small cell lung cancer-related fatigue and cachexia.                                         | The results are not yet posted.                                                                                                                                                           |
| PF-423691                | Crohn's disease                             | Completed/<br>NCT01345318                        | Pfizer                         | B0151005 open-label extension study (ANDANTE II).                                                                                                                 | PF-04236921, at 50 mg dose, induced clinical response and remission in moderate to severe Crohn's disease patients. <sup>108</sup>                                                        |
|                          | Rheumatoid arthritis                        | Completed/<br>NCT00838565                        | Pfizer                         | Phase I study of the safety, tolerability, pharmacokinetics and pharmacodynamics of multiple intravenously administered doses of PF-04236921 in patients with RA. | The current model used in the study enables simulation of PF-04236921 pharmacokinetics and pharmacodynamic profiles under various dosing regimens and patient populations. <sup>109</sup> |

<sup>a</sup> Status as on 01-10-2019 (date of accession).<sup>b</sup> Accessed from <https://clinicaltrials.gov>.<sup>110</sup>

these mAbs have achieved tremendous commercial success also, but high cost, invasive route of administration and high rate of immunogenicity remain their major limitations. Therefore, there is a need for the development of new IL-6 inhibitors having low antigenicity and being patient-friendly. Several synthetic and naturally occurring molecules are reported in literature as IL-6 inhibitors. These act by interfering with various routes involved in the production and signaling pathways of IL-6 (Fig. 4). These molecules have been tested by using lipopolysaccharide (LPS)-induced IL-6 mechanisms. LPS (an endotoxin) is the principal component of the outer membrane of gram-negative bacteria. Human cells are exquisitely sensitive to LPS, where it initiates various biochemical pathways leading to the expression of various inflammatory cytokines. It acts by binding with transmembrane signaling receptor *i.e.*, toll-like receptor 4 (TLR4) and its accessory protein MD2 present on the cell surface. These receptors activate various intracellular signaling pathways that include c-Jun N-terminal kinase (JNK), extracellular signal-regulated kinases (ERK) and p38, and also activate IL-6, which subsequently initiate various mechanisms as stated earlier in Section 5 and Fig. 3.<sup>111</sup> A comprehensive and systematic compilation of molecules acting on various steps of these pathways has been outlined (Fig. 4) that may assist medicinal chemists to develop new anti-IL-6 molecules acting through specific mechanisms.

### 6.1. IL-6 production inhibitors

Numerous compounds reported in several research reports are termed as anti-IL-6 on the basis of their ability to decrease the LPS-induced IL-6 levels in different cell lines. However, the precise mechanisms responsible for this decrease in levels of IL-6 remain unknown. Herein, all such molecules are classified as IL-6 production inhibitors.

One of the earliest reports on the evaluation of an in-house library of compounds as IL-6 inhibitors disclosed compound 1 (Fig. 5) as a potent IL-6 inhibitor (IC<sub>50</sub> 0.2 μM). Taking this as lead, Tagat et al. have developed a series of 2,3,7,8-tetrahydro-4-aryl-1*H*-cyclopent[*e*]imidazo-[1,2-*a*]-pyridin-5(6*H*)-one derivatives (1, Fig. 5) that decrease the levels of IL-6 in murine myelomonocytic leukaemia (WEH1-265.1) cells. The SAR studies have revealed that expansion of cyclopentane ring (A) to a cyclohexane or cycloheptane ring, or replacement of imidazole ring (C) with pyrimidine caused a complete loss of the activity. Replacement of -NH- in the imidazole ring with -N(CH<sub>3</sub>)-, -S- or -O- did not produce any favourable effects on the activity. Further modifications such as (i) different substituents on the phenyl ring, (ii) a spacer group between phenyl ring and the tricyclic core, and (iii) replacement of phenyl ring with a heteroaryl or other moieties revealed that introduction of a sulphur atom between the phenyl ring and tricyclic core, and replacement of the phenyl ring with any tested moiety result in complete loss of the activity. It proves that phenyl ring is an important structural component for IL-6 inhibition. A substituent at *ortho*-position of this phenyl ring increases the activity to greater extent than that at *meta*-position with an exception of -COOC<sub>2</sub>H<sub>5</sub> group, which incurs higher potency at *meta*-position. Substituents like -NO<sub>2</sub> or -NHCOCH<sub>3</sub> at *ortho*-position produce equipotent compounds.<sup>112</sup>

The same research group has developed and evaluated several derivatives of diaryl pyridine, diaryl pyrazine, and *meta*-terphenyl (2–4, respectively, Fig. 5) as IL-6 production inhibitors. Among these, diaryl pyrazine derivatives (3) exhibit the minimum IL-6 inhibitory activity. Any increase in the polarity by introduction of -Cl or -CN group in the central heterocyclic ring, decreases the activity. Placement of aromatic rings at 3- and 5-positions of the central pyridine ring, yields the most active pharmacophore *i.e.*, 3,5-diarylpyridine. Further, a polar group on these aryl rings tends to decrease the activity in the order of -COOH > -OH ~ -CH<sub>2</sub>OH > -CH = N-OH > -OCH<sub>3</sub> ~ -H. These substituents are best suited for the activity, when placed at *para*-position on the aryl rings. From this library of compounds, the analogs of 2 and 4 that bear hydroxyl groups at *para*-positions of both the phenyl

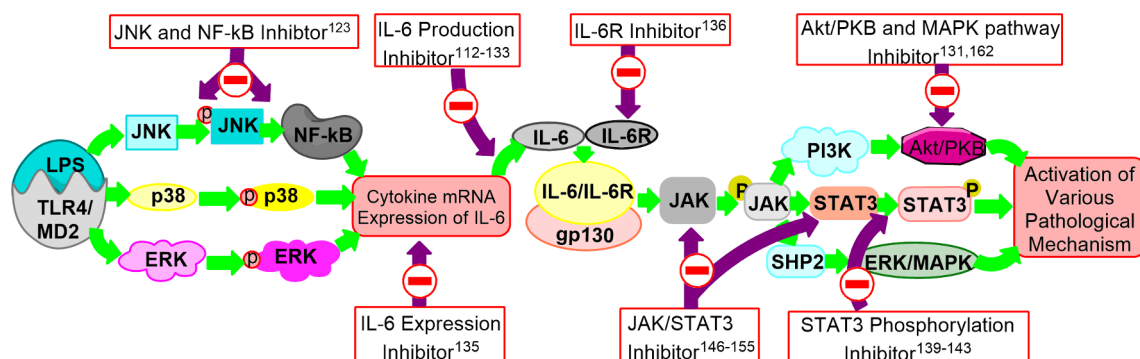


Fig. 4. Different routes of production and signaling pathways of IL-6. The encircled red bold line indicates the site  $\ominus$  at which IL-6 inhibitors act. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

**1**

| X                           | R                                      | % Inhibition <sup>a</sup> |
|-----------------------------|----------------------------------------|---------------------------|
| -NH-                        | H                                      | 74                        |
| N(CH <sub>3</sub> ) / S / O | H                                      | 43/ 52/ 20                |
| -NH-                        | -OCH <sub>3</sub> / -Cl                | 82/ 90                    |
| -NH-                        | -NO <sub>2</sub> / NHCOCH <sub>3</sub> | 73/74                     |
| -NH-                        | -NH <sub>2</sub>                       | 82                        |

**2** (X=N; Y=CH)  
**3** (X,Y=N)  
**4** (X,Y=CH)

| R                                 | % Inhibition <sup>a</sup> |    |           |
|-----------------------------------|---------------------------|----|-----------|
|                                   | 2                         | 3  | 4         |
| -OH                               | 73±7                      | 55 | 85±11     |
| -CH <sub>2</sub> OH               | 68±8                      | -  | 62        |
| -COOH                             | 84±9                      | -  | 50        |
| -CH=N-OH                          | 70±5                      | -  | 46        |
| -CH <sub>2</sub> NCH <sub>3</sub> | 60±0                      | -  | cytotoxic |

Fig. 5. Pioneer analogs (1–4) acting as IL-6 production inhibitors. <sup>a</sup>Percent inhibition reported at 10  $\mu$ M concentration.

rings show the maximal activity *in vitro* in LPS-induced IL-6 model ( $IC_{50}$  0.4 and 1.5  $\mu$ M, respectively). The same analogs, when administered orally, exhibit 71% and 88% inhibition of IL-6 production in mice serum at a dose of 25 mg/kg.<sup>113</sup>

(+)-Madindoline A and B (5, Fig. 6) are nontoxic stereoisomeric forms of isoindoline and are isolated from *Streptomyces nitrosporeus* K93-0711 by fermentation. These demonstrate dose-dependent inhibition of IL-6-induced growth of MH60 cell line. However, the yield of madindoline by fermentation process is very low.<sup>114,115</sup> Therefore,

Hosokawa and colleagues developed a novel synthetic method to generate the madindoline skeleton.<sup>116</sup> In 2005, Seleh *et al.* demonstrated that Madindoline A binds to gp130, specifically with D1 domain non-covalently, and inhibits IL-6-dependent signaling pathways.<sup>117</sup> Bharate *et al.* have designed and synthesized three series of hybrid compounds by condensation of 5,6-dihydro-4H-pyran-3-carbaldehyde, 2-formyl-3,4,6-tri-O-methyl-D-glucal or chromone-3-carbaldehyde with varied heteroaromatic rings. All compounds (6, Fig. 6) were evaluated against LPS-induced IL-6 production in THP-1 cell line and for cytotoxicity in CCK-8 cell line. All glucal-heterocycle, pyran-heterocycle and chromone-heterocycle hybrid analogues inhibited IL-6 by 35–44%, 19–66% and 9–60%, respectively at 10  $\mu$ M concentration. None of these hybrids, except chromone hybrids of pyrazolo-pyridine and benzimidazo-pyrimidine, exhibited any cytotoxicity at 10  $\mu$ M. The chromone-iso-xazolopyridine hybrids (6c1) displayed the most promising IL-6 inhibitory activity ( $IC_{50}$  0.16  $\mu$ M) from this series.<sup>118</sup>

A series of 1-benzyl-2,4,5-triphenyl-4,5-dihydro-1H-imidazole derived compounds (7, Fig. 7), bearing ethyl ester, methyl ester, benzyl ester, amide or alcoholic groups at 4-position, has demonstrated a strong dose-dependent inhibition of NF- $\kappa$ B-mediated IL-6 production. Among these, the ethyl ester derivative is the most potent ( $IC_{50}$  0.8  $\mu$ M) and is stable to blood esterase activity up to 24 h at 37° C, whereas the amide- and alcohol-substituted compounds are inactive. Furthermore, the *R,R*-enantiomer of the ethyl ester derivative is more potent ( $IC_{50}$  0.2  $\mu$ M) than the *S,S*-enantiomer ( $IC_{50}$  0.8  $\mu$ M).<sup>119</sup> The success of this pilot study led to further investigations on ethyl ester derivative by systematic exploring of varied substituents at  $R_1$ ,  $R_2$ ,  $R_3$  and  $R_4$  positions (8, Fig. 7). Exploration of  $R_1$  domain with CH<sub>3</sub>, *i*-C<sub>3</sub>H<sub>7</sub>, phenyl, benzyl and indolyl-3-methyl revealed that a phenyl group is most conducive to the IL-6 inhibitory activity. With phenyl ring placed at  $R_1$ , exploration of different esteric and amidic functionalities at  $R_2$  revealed that an ester group afforded analogues more active than the corresponding amide analogs. Among various esteric groups, an ethyl ester

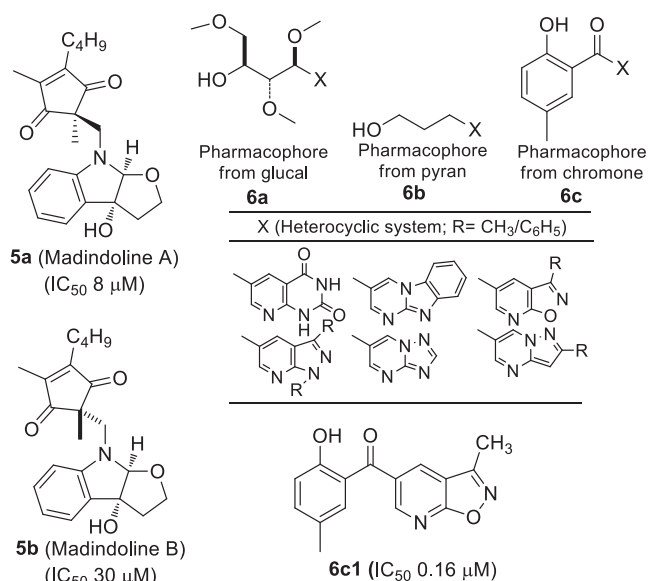
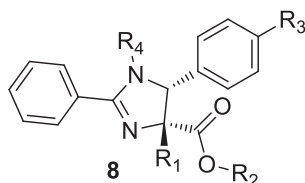
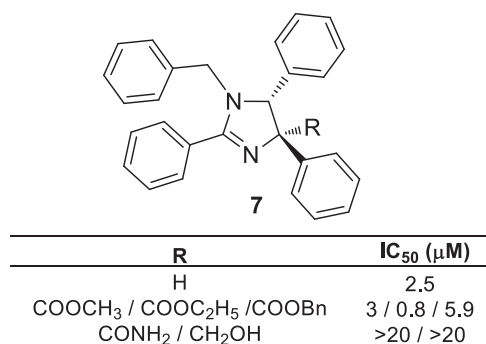


Fig. 6. Madindoline (5) and heterocyclic analog (6) as IL-6 production inhibitors.



| R <sub>1</sub>                                            | R <sub>2</sub>                                                          | R <sub>3</sub>                    | R <sub>4</sub> | IC <sub>50</sub> (μM) |
|-----------------------------------------------------------|-------------------------------------------------------------------------|-----------------------------------|----------------|-----------------------|
| CH <sub>3</sub> / <i>i</i> -C <sub>3</sub> H <sub>7</sub> | CH <sub>3</sub>                                                         | H                                 | Bn             | 3 / 7.1               |
| Ph / Bn                                                   | CH <sub>3</sub>                                                         | H                                 | Bn             | 3 / 4.1               |
| Ph                                                        | C <sub>2</sub> H <sub>5</sub> / <i>n</i> -C <sub>3</sub> H <sub>7</sub> | H                                 | Bn             | 0.8 / 1.9             |
| Ph                                                        | Bn                                                                      | H                                 | Bn             | 5.9                   |
| Ph                                                        | NH <sub>2</sub>                                                         | H                                 | Bn             | >20                   |
| Ph                                                        | C <sub>2</sub> H <sub>5</sub>                                           | Cl/F                              | Bn             | 0.3 / 1               |
| Ph                                                        | C <sub>2</sub> H <sub>5</sub>                                           | OCH <sub>3</sub> /CF <sub>3</sub> | Bn             | 1.3 / 0.8             |
| Ph                                                        | C <sub>2</sub> H <sub>5</sub>                                           | H                                 | Ac/Bz/Tos      | >20 / 6.6 / 6.6       |
| Ph                                                        | C <sub>2</sub> H <sub>5</sub>                                           | H                                 | 4-(F/Cl/Br)-Bn | 1.2 / 1.6 / 0.5       |

Fig. 7. Imidazole analogs (7–8) as IL-6 production inhibitors.

exhibits better activity than methyl and propyl esters. Thenceforth, with phenyl at R<sub>1</sub> and an ethyl ester at R<sub>2</sub>, R<sub>3</sub> domain was explored with various groups. The activity data revealed that a chlorine atom at R<sub>3</sub> is even more potent (IC<sub>50</sub> 0.3 μM) than ethyl ester analog of the lead compound 7. Finally, with a phenyl ring at R<sub>1</sub>, an ethyl ester at R<sub>2</sub> and a chlorine at R<sub>3</sub>, exploration of different *para*-substituted benzyl, benzoyl, imidazole, acyl and tosyl moieties at R<sub>4</sub> revealed that only halogenated-benzyl moiety at R<sub>4</sub> affords biologically active compound.<sup>120</sup>

Bandgar *et al.* have developed three series of compounds that exhibit promising inhibitory activity against LPS-induced IL-6 production in THP-1 cell line.<sup>121–123</sup> SAR studies on the first series of 2-hydroxy-4,6-dimethoxybenzophenones (9, Fig. 8) revealed that *N*-methylpiperazine/pyrrolidine at C-3 of ring A is most conducive to IL-6 inhibitory activity. Further, a chlorine or fluorine group at *meta*-position of the phenyl ring B yields IL-6 inhibitor 2–3 times better than those at the *ortho*-position. However, these derivatives have severe toxicity concerns as compared to the corresponding fluorine-substituted compounds. Hence, the compound with a fluorine at *meta*-position of B, and a *N*-methyl piperazine at C-3 of A is observed to be an effective inhibitor of IL-6 (97% inhibition at 10 μM concentration).<sup>121</sup>

All compounds of the second series, which was derived by placing methyl, methoxyl or halogens on phenyl ring B of 3-bromo-2-hydroxy-4,6-dimethoxybenzophenone (10, Figure 8), inhibited the production of IL-6 by 63–82% at 10 μM concentration. An electron-donating group that includes methyl or methoxyl, on the ring B increases anti-IL-6 activity as such, but triggers the cytotoxicity. Thus, in this series, *para*-fluoro substituted analog (63% inhibition at 10 μM concentration of IL-6) is preferred as a lead molecule over the *para*-methoxyl analog (82% inhibition at 10 μM concentration).<sup>122</sup> The third series is derived from the coupling of 2-hydroxy-4,6-dimethoxybenzoyl with  $\beta$ -chloro vinyl-chalcones (11, Figure 8). Exploration of varied electron-withdrawing and -releasing groups at various position of phenyl ring B revealed that these substituents, at *para*-position, are the most favourable to IL-6

inhibitory activity. The activity data revealed that the activity decreases as halogen is changed from fluorine to chlorine to bromine at *para*-position. This finding suggests that a more electronegative group on ring B incurs a higher IL-6 inhibitory activity. Also, two methoxyl groups, at either positions on the ring B, yield less potent compound than single methoxyl at *para*-position.<sup>123</sup>

Screening of another series of chalcone-based compounds (12, Fig. 8) for LPS-induced IL-6 inhibition in macrophages demonstrated that electron-donating substituent(s) in ring A and electron-withdrawing group(s) in ring B are favourable for the IL-6 inhibitory effect. The quantitative structure-activity data suggested that substituents on ring A are preferred in the order of 3,4,5-(OCH<sub>3</sub>)<sub>3</sub> > 3,4-(OCH<sub>3</sub>)<sub>2</sub> > 4-OCH<sub>3</sub> > 4-NH<sub>2</sub> and 5-NH<sub>2</sub> > 2-OH > H > 3,5-(F)<sub>2</sub>, whereas the presence of a halogen on ring B markedly increases the inhibitory activity. The further mechanistic studies suggested that compounds 12a and 12b interfere with JNK/NF-κB signaling mechanism by inhibiting the phosphorylation of JNK and p38 in a dose-dependent manner.<sup>124</sup>

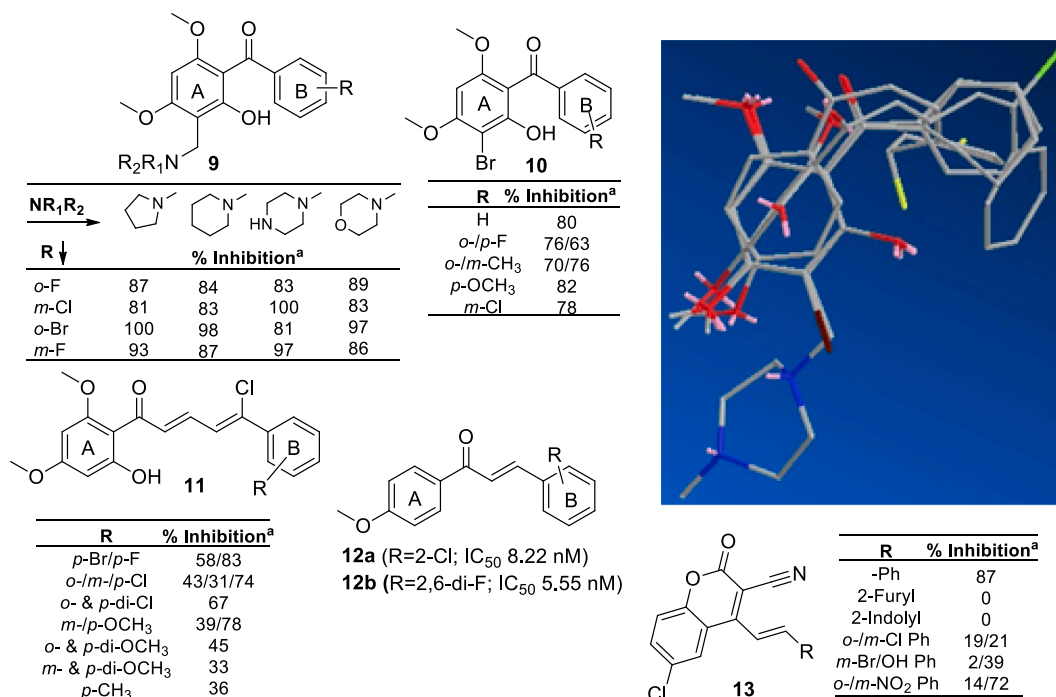
A critical analysis of these studies led us to propose that an increase in electron density at C-3 of ring A in series 9 and 10 increases the activity, but an increase in electron density in ring B causes a decrease in the activity. This proposition is based on the evidence of an increased potency by fluorine in all the three series. Further, it can also be suggested that an increase in length of chain between the two rings (A and B) in chalcones decreases the activity (from 63 to 100% to 31–83% inhibition at 10 μM concentration of IL-6). A 3D overlay of energy-minimised conformations, generated in ChemBio3D 12.0, of the most potent compounds from each of these series demonstrated that the ring A, ring B and the carbonyl group, in all three series of compounds, occupy the similar 3D-orientations, and all the substituents attached to rings A and B also occupy the same 3D space (Fig. 8). A mechanistic study on compound 12 suggested that it acts by inhibiting the JAK phosphorylation and interfering with NF-κB signaling mechanism. Based on the similarities in structures and 3D-orientations of compounds 9–12, it may be implied that compounds 9–11 also act through inhibition of JAK phosphorylation and NF-κB signaling to decrease the level of IL-6.

Upadhyay *et al.* synthesised a series of 4-styryl-3-cyanocoumarin derivatives (13, Fig. 8), which are conformationally similar to preceding chalcone derivatives, and evaluated them against LPS-induced IL-6 in THP-1 cell line. The SAR study unveiled that replacement of phenyl ring with furyl or indolyl yields inactive compounds, which in turn, demonstrate the importance of phenyl ring for the activity. Any substitution on this phenyl ring decreases the activity, which suggested the compound having unsubstituted phenyl ring as lead (87% inhibition of IL-6 at 10 μM). Further, substitution with a halogens (Cl, F, or Br) at *ortho* or *meta*-position on the phenyl ring yields less active compounds (4–21% inhibition at 10 μM concentration) as compared to that with nitro or hydroxyl (39–73% inhibition at 10 μM concentration).<sup>125</sup> These findings are contradictory to the preceding reports on chalcones, where a halogen atom on the same location affords more active compound.

Taking thalidomide as lead, Chaulet *et al.* designed imidazo-pyrrolopyridine-dione analogs (14, Fig. 9) that are more rigid than thalidomide, and 7-azaindole derivatives that are more flexible than thalidomide as IL-6 inhibitors. The activity data states that rigid analogs have no inhibitory effect on IL-6 production and TNF-α secretion. Among these, only compound 14a has exhibited significant inhibition of TNF-α and IL-6 at 100 μM concentration. In contrast, the flexible group of compounds, prototyped by 14b, elicit a dose-dependent inhibition of both IL-6 and TNF-α. Compound 14b emerged as the best inhibitor of IL-6 production from the series and better than thalidomide itself. The findings of this study suggested that a sulfonyl group, in less rigid molecules, can act as an important scaffold for designing anti-IL-6 agents.<sup>126</sup>

Two other series of rigid compounds developed from tricyclic systems, triazo-[4,3-*a*]-quinoxaline and benzo[*g*]pteridines (15 and 16,

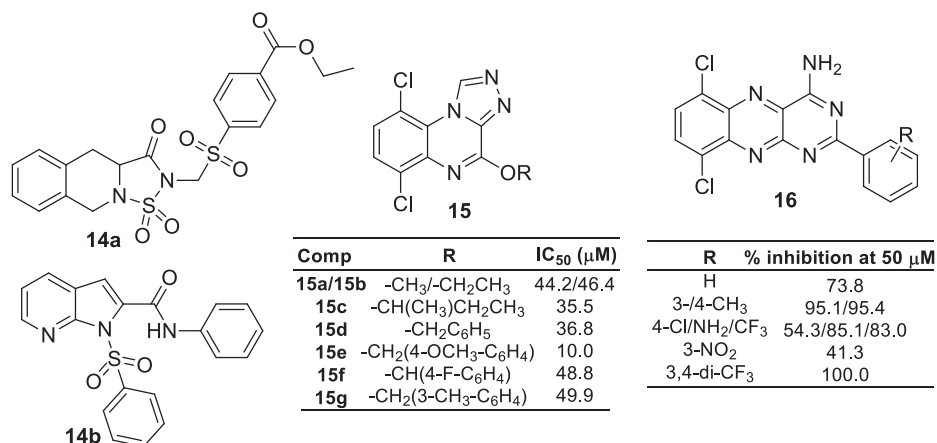




**Fig. 8.** Benzophenone (9–10), chalcones (11–12) and 4-styryl-3-cyanocoumarin (13) derivatives as IL-6 production inhibitors; and 3D representation of overlaid most potent compounds among 9, 10, 11 and 12b series. <sup>a</sup>Percent inhibition reported at 10  $\mu$ M concentration.

respectively, Fig. 9) are evaluated for inhibitory activity against LPS-induced IL-6 production in human myeloid cell line HL-60, and for cytotoxicity at 50  $\mu$ M.<sup>127,128</sup> The SAR studies involving varied alkoxy groups at 4-position of 6,9-dichloro-1,2,4-triazo-[4,3-*a*]-quinoxaline (15) revealed that the activity increases with increase in the size of aliphatic group. Compound 15c, bearing the bulkiest aliphatic group in the study, is maximally potent (IC<sub>50</sub> 35.5  $\mu$ M). Replacement of this aliphatic group with benzyl group retains the activity (15d, IC<sub>50</sub> 36.8  $\mu$ M). Further, among the various substituents (F, CH<sub>3</sub>, OCH<sub>3</sub>) at *para*-position of benzylic phenyl ring, the methoxyl group afforded the most potent compound (15e, IC<sub>50</sub> 10.0  $\mu$ M).<sup>127</sup> A structure-activity data generated by exploring varied substituents at *ortho*-, *meta*- and *para*-positions on the phenyl ring in 2-substituted-phenyl-4-amino-6,9-dichlorobenzo[g]pteridines (16) revealed that a methyl or a trifluoromethyl group yields the maximally potent compound. Both *meta*-methyl and *para*-methyl substituted analogs are found equipotent (95% inhibition at 50  $\mu$ M concentration), whereas trifluoromethyl group at *meta*-position afforded 100% inhibition at 50  $\mu$ M concentration IL-6 inhibition with IC<sub>50</sub> of 22.9  $\mu$ M.<sup>128</sup>

Exploration of variedly substituted alkyl groups around indole and indoline nuclei (17 and 18, respectively, Figure 10) for inhibitory potential against LPS-induced production of IL-6 in RAW-264.7 cell line has divulged that, irrespective of the substituent on the benzo ring, a 3-substituted indoline derivative (18) is significantly more potent than the corresponding 1-substituted indole derivative (17). Compounds 18a and 18b (bearing methyl ester of propionyl on the benzo ring) reduced the IL-6 levels significantly, whereas the compounds having 2-(dimethylamino)ethyl group on the benzo ring (18d and 18e) are least active against the production of IL-6. Further investigations for signaling mechanism using 18a, which is the most potent IL-6 inhibitor from the series, suggest that it inhibits the phosphorylation of p38 and reduces the levels of NF- $\kappa$ B. All indoline derivatives (18a–18e) are also found to be AChE inhibitors with IC<sub>50</sub> ranging from 0.4 to 5.5  $\mu$ M. This dual nature of these 3-substituted indoline derivatives makes them suitable drug candidates for the treatment of complex neurodegenerative disorders such as AD and Parkinson's disease.<sup>129</sup> Other series of indole-based compounds include indole-2-aminophenylmorpholin-3"-ones with benzoyl or benzyl group at N-1 of indole nucleus. (19 and 20,



**Fig. 9.** Tricyclic derivatives (14–16) as IL-6 production inhibitors.

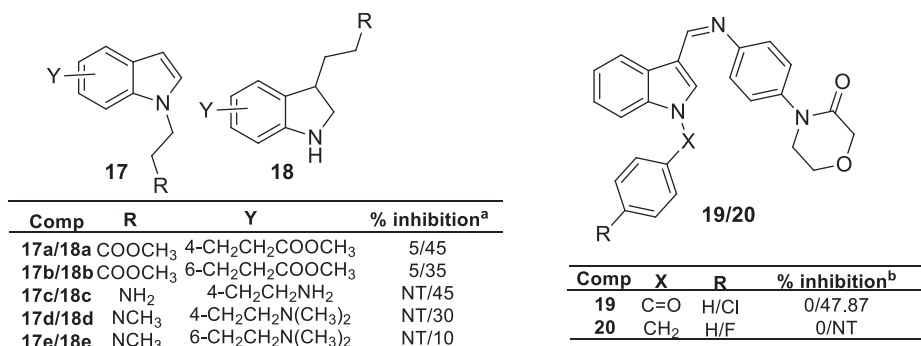


Fig. 10. Indoline derivatives (17–20) as IL-6 production inhibitors. <sup>a</sup>Percent inhibition w.r.t. LPS control reported at 10 nM concentration, <sup>b</sup>Percent inhibition w.r.t. LPS control reported at 10  $\mu$ M concentration, NT- not tested.

Figure 10). Out of these, 4-substituted N-benzoyl (19)/-benzyl (20) derivatives inhibit IL-6 better than the unsubstituted derivative in microglial cell (BV2 cell line). The *para*-chloro derivative (19, R = Cl) is proven to be the most potent IL-6 inhibitor (70% inhibition at 10  $\mu$ M) from the series. In addition, it also inhibits the nuclear translocation of NF- $\kappa$ B and AP-1.<sup>130</sup>

Plants have remained the primary sources of lead therapeutic agents for treatment of any disease or disorder. Many chemicals isolated from plants are also explored for their IL-6 production inhibitory potential. Andrographolide (21a, Fig. 11), isolated from leaves of *Andrographis paniculata* inhibits the expression of cytokine in murine macrophages via suppression of ERK1/2 and PI3K-Akt signaling pathways. The hydroxyl group on  $\gamma$ -lactone ring of andrographolide makes it chemically unstable. Therefore, Li *et al.* have prepared some derivatives of andrographolide and evaluated them against LPS-induced IL-6 release in mouse macrophages (J774a.1 cell line). Among these, esteric derivatives (21c) are more potent as compared to the tricyclic derivatives (21b). The phenyl esters in 21c exhibit better inhibitory activities than the alkyl esters. A -NO<sub>2</sub> or a -CH<sub>3</sub> group on the phenyl ring at *para*-position further improves the IL-6 inhibition.<sup>131</sup>

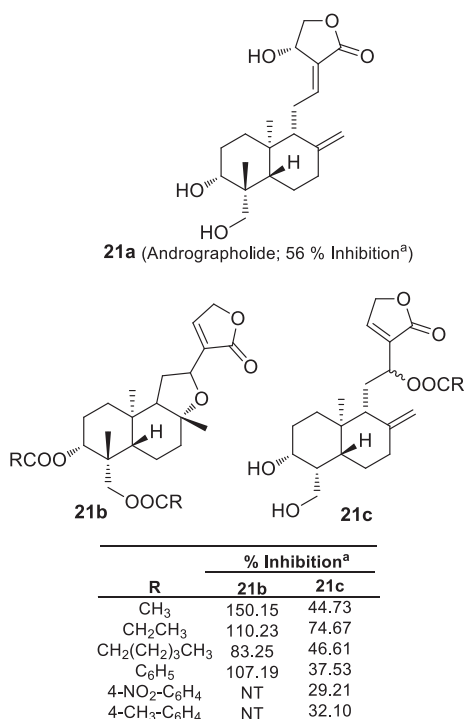


Fig. 11. Andrographolide and its derivatives (21) as IL-6 production inhibitors. <sup>a</sup>Percent inhibition reported at 10  $\mu$ M concentration. NT- not tested.

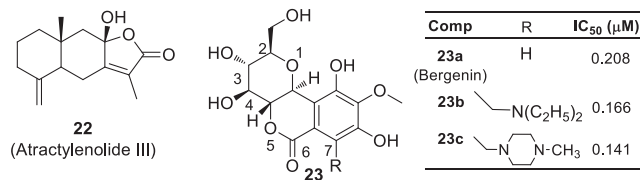


Fig. 12. Atractylenolide III (22) and Bergenin derivative (23) as IL-6 production inhibitors.

Atractylenolide III (22, Fig. 12) is the major bioactive component of *Atractylodes lancea*, and is reported to decrease the secretion of IL-6 by inhibiting the activation of p38 MAPK, C-Jun-N-terminal protein kinase, and NF- $\kappa$ B.<sup>132</sup> Bergenin (23, Fig. 12), a fused C-glycoside isolated from ethanolic extract of whole plant of *Bergenian ciliate*, is identified as a potent IL-6 inhibitor (IC<sub>50</sub> 0.208  $\mu$ M). Structural modifications at its C-7 by electrophilic substitution with various secondary amines has produced several compounds, all of which are found to be potent IL-6 inhibitors (IC<sub>50</sub> 0.14–1.63  $\mu$ M). The dimethylamine (23a) and *N*-methylpiperazine (23b) substituted analogs emerged as the most potent inhibitors with IC<sub>50</sub> 0.166 and 0.141  $\mu$ M, respectively. These derivatives also display good *in vivo* efficacy in collagen-induced arthritis model as manifested by decrease in swelling of paw and in IL-6 levels. These compounds are also observed to inhibit the expression of NF- $\kappa$ B significantly.<sup>133</sup>

After an extensive and critical analysis of these varied IL-6 production inhibitors reported in literature, we could identify, broadly, two structural components important for a compound to be useful as IL-6 production inhibitor. These are 2-hydroxy-4,6-dimethoxyacetophenone (also known as xanthoxylin) and an appropriately substituted fused heteronuclear moiety (Fig. 13). Xanthoxylin is reported to be a phytoconstituent in many plants that are utilized for their anti-inflammatory activities,<sup>134</sup> but no structure-activity co-relation has been suggested so far. However, the significant activity of xanthoxylin-based or -derived synthetic IL-6 inhibitors (such as compounds 6 and 9–12) suggest that anti-inflammatory activity of xanthoxylin-rich plants may be a function of xanthoxylin itself. The functional groups, which are most influential for incurring IL-6 inhibitory activity, around the heteronuclear moiety like indole, indoline and benzoxazole are summarized and depicted in Fig. 13. While, different substituents at N-1, C-2, and C-3 are widely explored, the positions C-5, C-6 and C-7 remain relatively unexplored.

## 6.2. IL-6 expression inhibitors

Production of IL-6 is mediated by IL-6 mRNA expression, which is regulated by various endogenous biomolecules under normal as well as pathological states. Interfering IL-6 mRNA expression has also been explored as another therapeutic strategy to curb an increase in levels of IL-6. Three steroids (24a–24c, Fig. 14), isolated from Korean marine

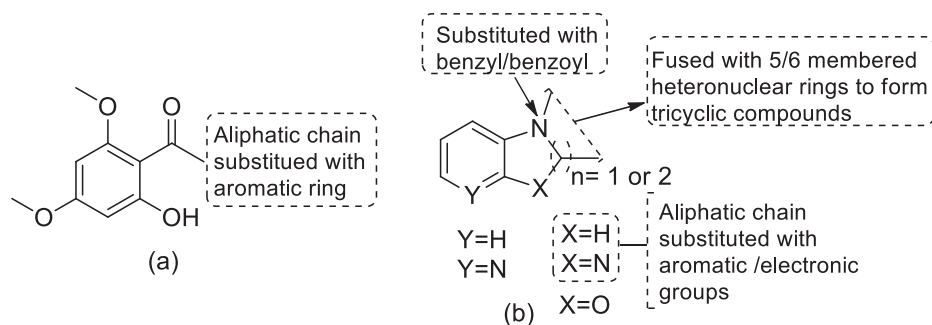


Fig. 13. Structural components important for incurring IL-6 production inhibitory activity.

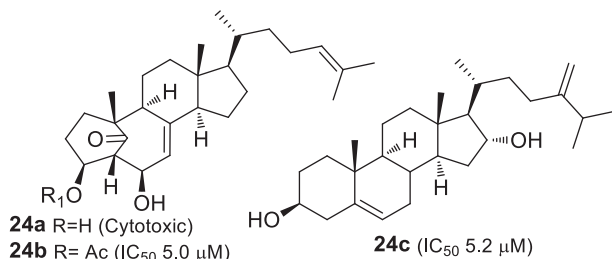


Fig. 14. Steroidal IL-6 expression inhibitors (**24**) derived from marine source.

organisms (a sponge of the genus *Monanchora*, class *Demospongiae*, order *Poecilosclerida*, family *Crambeidae*) are evaluated for their anti-inflammatory activity through quantitative influence on IL-6 mRNA expression in LPS-stimulated murine RAW264.7 macrophage cells by real-time quantitative PCR (RT-qPCR). Compound **24a** is found to be cytotoxic against the cells, and hence, not investigated further. Compounds **24b**, which differs from **24a** simply by acetylation at C-3, and **24c** are devoid of cytotoxicity and exhibit potent inhibitory effect on IL-6 mRNA expression in a dose-dependent manner with  $IC_{50}$  of 5.0 and 5.2  $\mu$ M, respectively.<sup>135</sup>

### 6.3. IL-6 receptor antagonists

Epoxyresibufogenin-3-formates are steroidal bufenolides isolated from Chinese toad skin extract.<sup>136</sup> These exist in two stereoisomeric forms i.e., 20*R*, 21*R*- and 20*S*, 21*S*-epoxyresibufogenin-3-formate (**25a** and **25b**, Fig. 15). An examination of their interactions with IL-6R suggested **25a** as the pharmacologically active diastereomer. In addition, it suppresses IL-6-induced  $\alpha$ 1-antichymotrypsin mRNA expression better than **25b**.<sup>137</sup>

### 6.4. IL-6/JAK/STAT3 signaling pathway inhibitors

The inflammatory IL-6/JAK/STAT3 signaling pathway contributes to various types of diseases as discussed earlier in Section 5. JAK2/

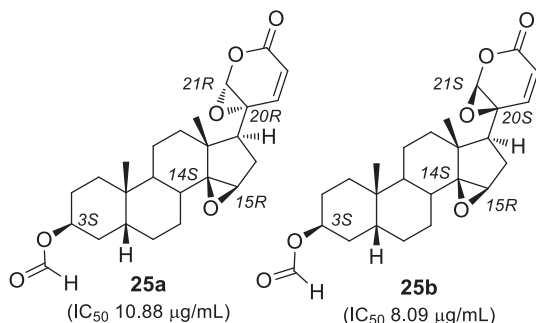


Fig. 15. Epoxyresibufogenin-3-formate (**25**) as IL-6R inhibitor.

STAT3 signaling pathway is constitutively activated. Strategies to suppress this pathway include (i) blocking the IL-6/IL-6R interactions; (ii) blocking the activation of JAK; (iii) blocking the phosphorylation of STAT3; and (iv) blocking the STAT3 DNA-binding activity.<sup>138</sup>

#### 6.4.1. STAT3 inhibitors

A library of 3-(pyrrolidinylsulfonyl)thiophene and 3-(piperidinylsulfonyl)thiophene derivatives (**26**, Fig. 16) has been designed, synthesized and evaluated for MTS cell viability assay using MDA-MB-231 breast (STAT3-dependent) and A4 (STAT3-null) cancer cells. From this library, only **26a** caused a 40% reduction in viability of MDA-MB-231 cells at 125  $\mu$ M concentration, and emerged as a potential STAT3-specific inhibitor. It inhibited the transcriptional activity of STAT3 ( $EC_{50}$  15  $\mu$ M) in HeLa cells that are treated with oncostatin M to activate STAT3 signaling via IL-6/gp130 receptor. Docking of **26a** in the crystal structure of IL-6/IL-6R/gp130 (PDB 1P9M) with two different sites of the protein, as depicted in Fig. 16, has revealed that the first site of its interaction is at the IL-6/IL-6R interface, and the second site is at IL-6/gp130 interface. It binds and prevents the interactions at both the sites.<sup>139</sup>

Among several saponins, isolated from aerial parts of *Gynostemma pentaphyllum*, evaluated for their inhibitory effect on IL-6-induced STAT3 activation in Hep3B cells, compounds **27–29** (Fig. 17) showed potent inhibitory effects ( $IC_{50}$  0.27, 0.59, and 0.50  $\mu$ M, respectively) in

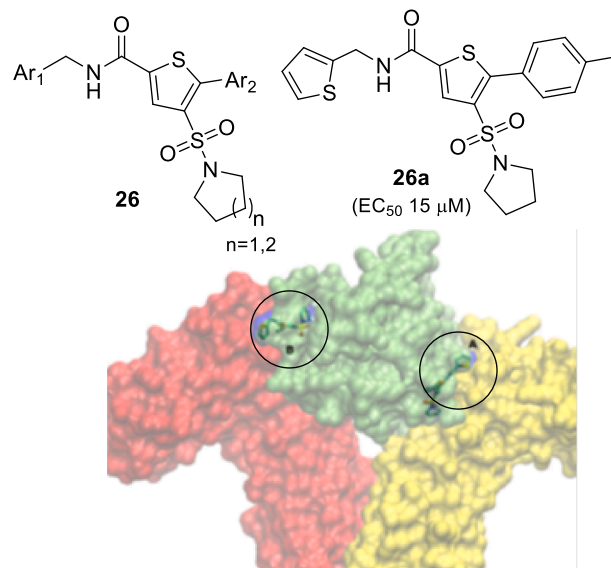


Fig. 16. Thiophene derivative (**26**) as IL-6-induced STAT3 inhibitors, and potential binding sites of **26a** at protein-protein interfaces of IL-6/IL-6R and IL-6/gp130.<sup>140</sup> Green colour represents IL-6, red colour represents IL-6R, and yellow colour represents gp130, Copyright 2019 Elsevier Ltd. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

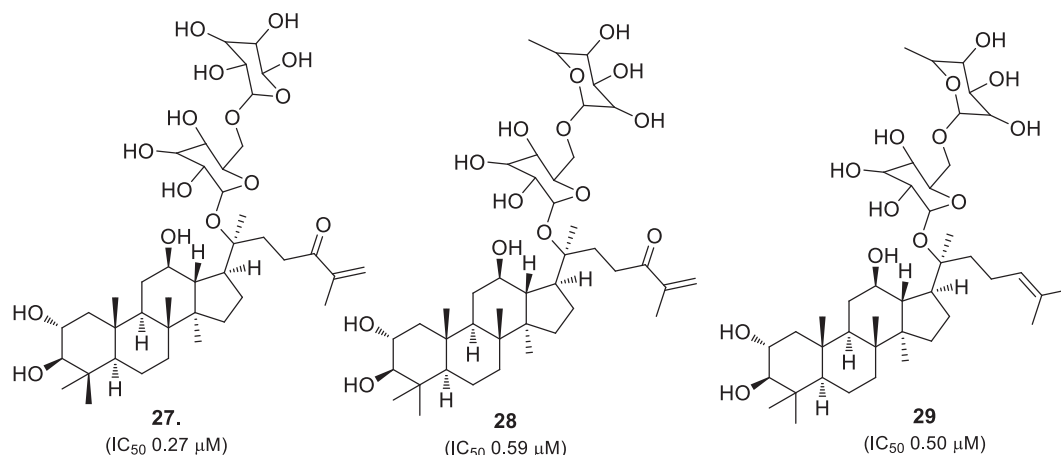


Fig. 17. Saponins (27–29) as IL-6-induced STAT3 inhibitors.

comparison to the positive controls *i.e.*, static (IC<sub>50</sub> 0.51  $\mu$ M) and genistein (IC<sub>50</sub> 15.1  $\mu$ M). Moreover, the aglycon portion of these compounds as such exhibited potent bioactivity (IC<sub>50</sub> < 0.6  $\mu$ M).<sup>140</sup>

Singh *et al.* have developed series of oxazolidine and indole derivatives (30 and 31, Fig. 18), and evaluated them for IL-6 signal blocking activity by luciferase expression in human hepatocarcinoma HepG2 cells transfected with p-STAT3-Luc. From the series, compound 30d is identified as a potent IL-6 signaling blocker (IC<sub>50</sub> 5.9  $\mu$ M). This compound is found even more potent than the standard inhibitor of IL-6 *i.e.*, (+)-Madindoline A (IC<sub>50</sub> 21  $\mu$ M). The SAR study has revealed that the activity increases with increasing alkyl chain up to *n*-pentyl. Further increase in alkyl chain length results in a decrease in the activity. The *R*-enantiomer of 30d does not show any inhibitory activity up to a concentration of 20  $\mu$ M, which suggests that *S*-(+)-oxazolidinone stereochemistry is required for appropriate interactions with the target site. Replacement of oxazolidinone ring with an indole ring yielded compounds 31, which show potent inhibitory activity against IL-6 as well as cytotoxicity. Computational studies-assisted mechanistic investigations revealed that compound 30d binds with hot-spot (Leu57 and Try157 of IL-6) similar to madindoline-A (Fig. 18). In addition, it also binds with

gp130 D1 domain of the hexameric complex through H-bonding with Val93 and Cys6 residue of the domain (Fig. 18).<sup>141</sup>

Pterostilbene (32, Fig. 19) and 3'-hydroxypterostilbene (33, Fig. 19), isolated from *Sphaerophysa salsula*, *Pterocarpus marsupium*, and honey bee propolis are evaluated for efficacy in azoxymethane (AOM)/dextran sodium sulfate (DSS)-induced colitis-associated cancer. Both 32 and 33 interfered with the AOM/DSS-induced shortening of colon and the increasing number of tumor cells. Both the compounds also significantly decreased the expression of IL-6. Dietary treatment of AOM/DSS-treated mice with 32 completely restored the reduced levels of SOCS3 protein, which is a negative regulator of IL-6/STAT3 signaling. It suggests that 32 might be acting *via* down-regulation of IL-6 signaling in the colonic tissue.<sup>142</sup>

Kim *et al.* have synthesized derivatives of benzoxazole (34, Fig. 19), and evaluated for the suppression of IL-6-induced phosphorylation of STAT3-mediated signaling in HepG 2 cells. The structure-activity data revealed that the compound having unsubstituted amino group show good inhibitory activity (97.5% inhibition at 20  $\mu$ g/ml concentration, IC<sub>50</sub> 18.9  $\mu$ M). Among the benzoyl and phenylacetyl derivatives of substituted amide, the benzoyl derivatives have better inhibitory

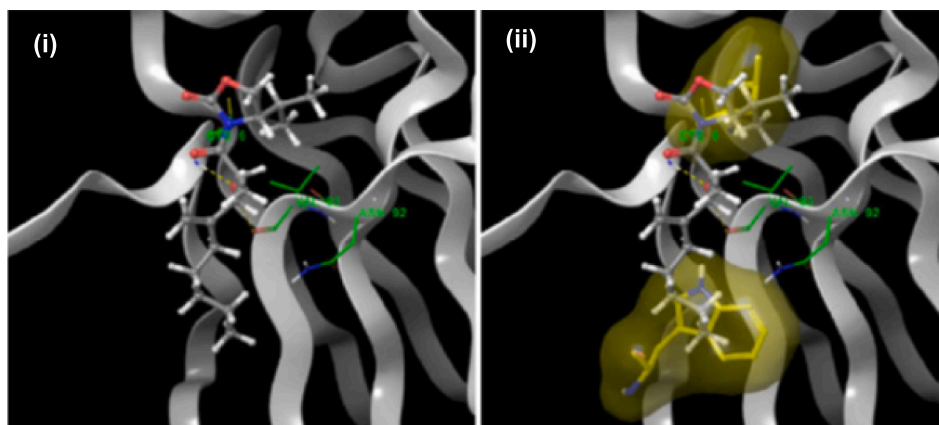
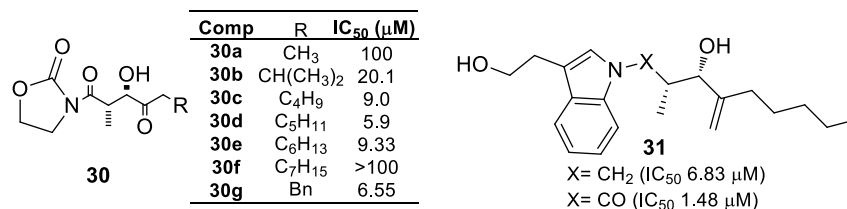


Fig. 18. Oxazolidine (30) and indole (31) derivatives as IL-6-induced STAT3 inhibitors. Compound 30d bound to gp130 D1 domain (i), and overlapping of 30d with Leu52 and Trp157 of IL-6 (ii).<sup>140</sup> Copyright 2019 Elsevier Ltd.



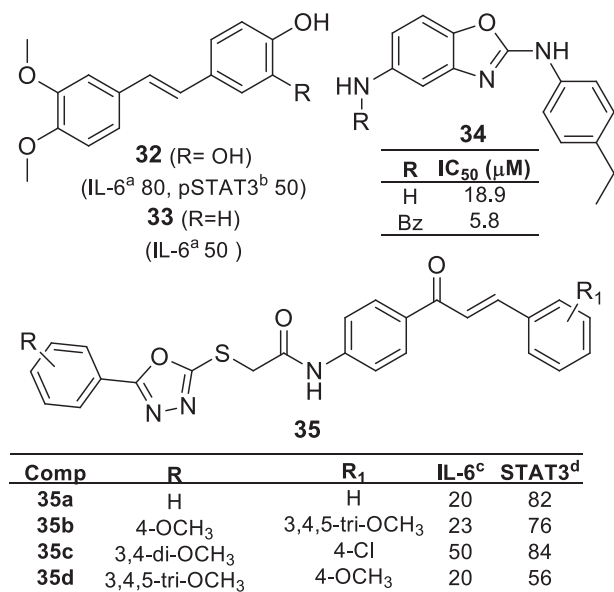


Fig. 19. Varied heterocyclic analogs (32–35) as IL-6-induced STAT3 inhibitors.

<sup>a</sup>Percent inhibition of IL-6 expression at a dose of 250 mg/kg, <sup>b</sup>Percent inhibition of pSTAT3 expression in AOM/DSS-induced IL-6 model at a dose of 50 mg/kg, <sup>c</sup>Percent IL-6 inhibition at 10  $\mu$ M, <sup>d</sup>Percent STAT3 inhibition in K-562 cells at 10  $\mu$ M.

activity (93.1% inhibition at 20  $\mu$ g/ml concentration, IC<sub>50</sub> 5.8  $\mu$ M). Further substitution an electron-withdrawing group (-Cl, -F, -CF<sub>3</sub>) at different positions of the benzoyl ring decreases the activity, whereas an electron-releasing group (-OCH<sub>3</sub>) increases the activity. These prototypical compounds *i.e.*, the compounds with unsubstituted amino group and unsubstituted benzoyl group are more potent than mandoline-A (IC<sub>50</sub> 20  $\mu$ M).<sup>143,144</sup> From a series of 1,3,4-oxadiazole-chalcone hybrids (35, Fig. 19), compounds 35a–35d exhibit potent cytotoxicity against different cancer cell lines. Compound 35a, 35b and 35d are also reported to down-regulate the IL-6 levels (20–23% inhibition at 10  $\mu$ M concentration), and inhibit the activation of STAT3.<sup>145</sup>

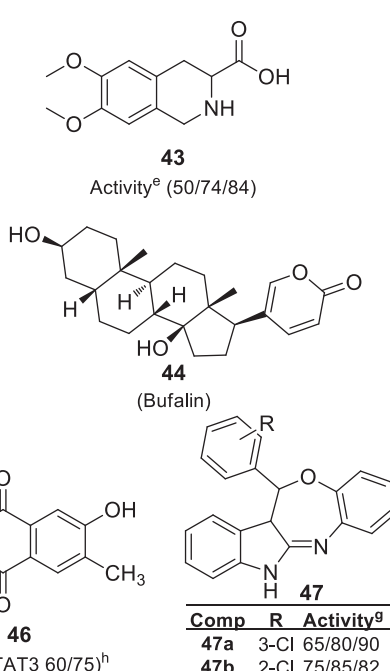
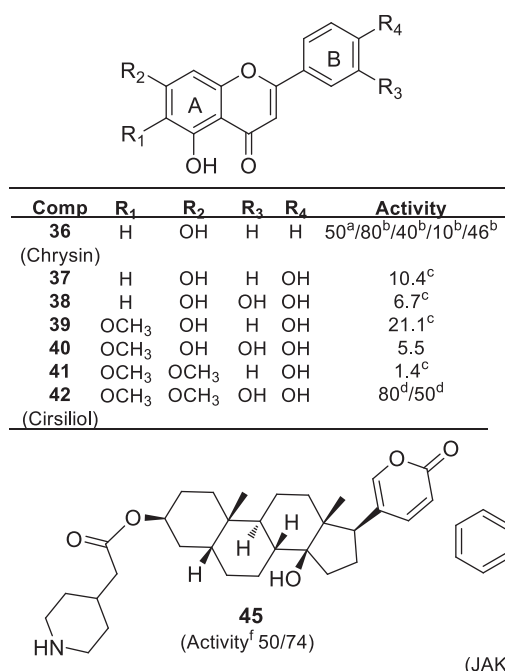
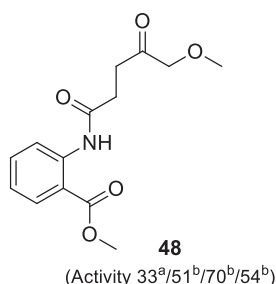


Fig. 20. Natural (36–46) and synthetic (47) IL-6-induced JAK/STAT3 inhibitors. <sup>a</sup>Percent decrease in vascular count for IL-6, <sup>b</sup>Percent decrease in protein levels of sIL-6R $\alpha$ /gp130/pJAK/pSTAT3, respectively at 10 nM concentration in comparison to 10 ng/ml of IL-6 stimulator; <sup>c</sup>IC<sub>50</sub> of STAT3; <sup>d</sup>Percent inhibition of phosphorylation of JAK2/STAT3, respectively at 20  $\mu$ M concentration; <sup>e</sup>Percent decrease in relative band density of IL-6/pJAK/pSTAT3, respectively at 25  $\mu$ M; <sup>f</sup>Percent decrease in relative protein level of pSTAT3/pJAK in comparison to STAT3/JAK, respectively at a dose of 2 mg/kg; <sup>g</sup>Percent decrease in relative band density of IL-6/pJAK/pSTAT3 at 10 mg/kg dose; <sup>h</sup>Percent decrease in protein levels on IL-6 induced JAK/STAT3 signaling in A549 cell at 80  $\mu$ M concentration.

#### 6.4.2. JAK/STAT3 inhibitors

Chrysin (5,7-dihydroxyflavone, 36, Fig. 20) is a flavonoid isolated from *Scutellaria baicalensis* Georgi, and is found to suppress the IL-6-induced neovascularization, human umbilical vein endothelial cell (HUVECs) migration and tube formation significantly in ovo of chicken. The molecular mechanistic studies *via* western blot method indicate that chrysin is able to down-regulate the expression of gp130, sIL-6R, phosphorylated JAK1 and STAT3, and VEGF in HUVECs. When the levels of STAT3 are knocked down, chrysin is not able to suppress the VEGF expression directly, which suggests that chrysin attenuate the levels of VEGF *via* STAT3. Thus, it is suggested that chrysin suppresses IL-6-induced angiogenesis through modulation of the sIL-6R/gp130/JAK1/STAT3 signaling pathway.<sup>146</sup> Out of twenty two known compounds isolated from *Salvia plebeian*, five flavonoids apigenin (37), luteolin (38), hispidulin (39), nepetin (40) and cirsimaritin (41) exhibited inhibitory effect against the IL-6-induced STAT3 activity. The SAR suggests that luteolin and nepetin, which have a 3,4-dihydroxy arrangement in B-ring, exhibit about 1.5 to 4-fold higher inhibitory activity in comparison to apigenin and hispidulin, which contain a monohydroxyl group in B-ring. Cirsimaritin, which has a 6,7-dimethoxyl group in A-ring, is more potent than hispidulin having one methoxyl group. These flavonoids also inhibit the phosphorylation of JAK2 and STAT3.<sup>147</sup> Taking these findings as lead, Lim *et al.* evaluated cirsiolol (42) for its potential against the IL-6-induced JAK/STAT3 signaling pathway. It inhibits not only the IL-6-induced phosphorylation of STAT3 and JAK2, but also decreases the phosphorylation of gp130 and ERK, and the IL-6-induced gene expression of IL-1 $\beta$ .<sup>148</sup> 6,7-Dimethoxy-1,2,3,4-tetrahydroisoquinoline-3-carboxylic acid (43, Fig. 20) is an alkaloids isolated from *Mucuna pruriens* seeds. Investigations on this molecule for its anti-proliferative activity against human hepatoma cells<sup>149</sup> and colorectal carcinoma (CRC)<sup>150</sup> revealed that it enhances the anti-oxidative physiological processes and causes the disappearance of infiltration of inflammatory cells, tubular adenoma, tumoral deposits and tumoral vacuoles. Additional studies involving gene expression analysis of IL-6 and Western blot analysis provided the evident roles of 43 in down-regulation of IL-6-induced activation and phosphorylation of JAK2 and STAT3. These findings suggest that 43 exhibits anti-CRC potential through the blockade of IL-6/JAK2/STAT3 oncogenic signaling.<sup>151</sup> Bufalin (44, Fig. 20) is another phytoconstituent isolated from Chinese toad, and is reported to produce significant cytotoxicity in



**Fig. 21.** Benzoate derivative (**48**) as MAPK, PI3k/Akt pathway inhibitor. <sup>a</sup>Percent decrease in relative mRNA expression of IL-6, <sup>b</sup>Percent decrease in levels of p-PI3k/p-Akt/pMAPK at 10  $\mu$ g/mL.

various malignancies. But its therapeutic utility is challenged due to its high cardiac toxicity, poor water solubility, rapid metabolism and other side effects.<sup>152,153</sup>

A modified analog of bufalin (**45**, Fig. 20) induces a greater apoptotic effect with lower acute toxicity. Among a battery of cancer cell lines, the multiple myeloma cell lines *i.e.*, ARP-1 and CAG are maximally sensitive to **45** (IC<sub>50</sub> 0.98 nM and 1.56 nM, respectively). Compound **45** has been reported to act by decreasing the expression of IL-6 mRNA. It is also reported to suppress the phosphorylation of JAK2 (Tyr 1007/1008) and STAT3 (Tyr 705), which inhibits the JAK2/STAT3 signaling pathway by attenuating IL-6 expression.<sup>154</sup> 2-Hydroxy-3-methylanthraquinone (**46**, Fig. 20) is a major anthraquinone component of a traditional Chinese plant *Hedyotis diffusa* Wild. It has been reported to inhibit several types of cancer cell lines.<sup>155–157</sup> Sun et al.<sup>158</sup> have found that it acts by inhibiting the IL-6-induced JAK2/STAT3 pathway in A549 lung cancer cell line. It inhibits IL-6-stimulated growth in cell, and also effectively down-regulates the phosphorylation of JAK2/STAT3.

A SAR study on varied indole-fused benzoxazepine derivatives (**47**, Fig. 20) involving different substituents at different positions of the phenyl ring demonstrated that compounds having a chlorine atom at *ortho* (**47a**) or *meta* (**47b**) position is most potent against Hep-G2 liver cancer cell line. *In silico* and molecular dynamic simulation studies prove compound **47b** to have good structural stability in the active site of IL-6.<sup>159</sup> Based on these observations, role of both the compounds, **47a** and **47b** in IL-6 signaling mechanism were investigated using *N*-Nitrosodiethylamine (NDEA)-induced hepatocellular carcinoma (HCC) rat model. These are found to down-regulate the level of IL-6. Suppression of IL-6 level with **47a** is more pronounced than that with **47b**. Compound **47a** also markedly decreases the expression of JAK2 and STAT3.<sup>160</sup>

Analysis of this literature on IL-6 signaling inhibitors *via* the inhibition of STAT3 has culminated in identification of two salient structural features in IL-6 inhibitor. Firstly, a di-substituted five-membered ring is preferred over a mono-substituted five-membered ring. The substituents may be long chain aliphatic groups or substituted aromatic rings. Secondly, a benzene ring fused with the five-membered ring increases the IL-6-induced STAT inhibition, but its cytotoxicity potential will be guided by the type of substituent(s) on it and the nature of the fused five-membered ring.

#### 6.5. MAPK, PI3k/Akt pathway inhibitor

IL-6 stimulates MAPK and Akt pathways, which further leads to the onset of various pathological conditions.<sup>161</sup> Methyl 2-(4'-methoxy-4-oxobutanamide)benzoate (**48**, Fig. 21), isolated from *Helianthus tuberosus* L. (which is also known as *Jerusalem srthoke*), has been found to suppress the LPS-induced secretion of IL-6 in RAW 264.7 cells. This compound is reported to reduce the phosphorylation of MAPKs without affecting its activation, and down-regulate the activation of PI3k and Akt in response to LPS. This decreased activation of PI3k/Akt, as expected, led to the suppression of I $\kappa$ B- $\alpha$  degradation.<sup>162</sup>

## 7. Conclusion and perspectives

IL-6 plays an important role in the physiology and pathology of various diseases through diverse mechanisms. The present review summarizes the literature on structure and function of IL-6 and its receptor. The major emphasis of this report is to provide a comprehensive data on compounds, of natural or synthetic origin, established as inhibitors of IL-6. Various research reports, since the first one in 1995, existing in literature on IL-6 inhibitors have been classified on the basis of their proposed or established mechanism of action *i.e.*, IL-6 production inhibitors or IL-6 signaling mechanism inhibitors. The signaling mechanism inhibitors are further sub-classified, depending on the enzyme or mediator inhibited en route IL-6 signaling pathway. Through a knowledge-based analysis of chemical moieties responsible for inhibition of IL-6 in each category has led to identification of the structural components that are critical and important for good IL-6 inhibitory activity. SAR and common salient features identified from these studies have been reported, wherever possible. This comprehensive analysis would be helpful to medicinal chemists to design, through knowledge-based and/or software-based approach(s), novel compounds capable of inhibiting IL-6 *via* either mechanism(s). Such IL-6 inhibitors may prove useful tool(s) in research on and treatment of complex diseases such as cancer, neurodegenerative disorder, IBD, RA, CD and asthma.

## Declaration of Competing Interest

There is no conflict of interest.

## Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.bmc.2020.115327>.

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